SYNTOMIC MORPHISMS ARE GORENSTEIN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0C15 (sources/stacks-duality/pinned-original-source.tex line 6559).

Allowed standard prerequisites (editorial).

- Syntomic morphisms are flat with lci fibres. Morphisms, Lemma labelled morphisms-lemma-syntomic-flat-fibres, cited at proof lines 6564-6568. Gives flatness and the fibre condition.
- A local complete intersection over a field is Gorenstein. Same chapter, Lemma labelled lemma-gorenstein-lci, cited at proof lines 6568-6570. Supplies the Gorenstein property fibre-wise.
- Syntomic equals flat and local complete intersection. More on Morphisms, Lemma labelled more-morphisms-lemma-flat-lci, cited at proof lines 6570-6573. Gives the equivalent formulation.

Source reference. Original lemma 01UF: sources/stacks-chow/tags.txt, source lines 2408–2408. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6567 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BVA: sources/stacks-duality/pinned-original-source.tex, source lines 6424–6424. Same-chapter pointer into the author environment lemma at source line 6424; invoked at proof line 6569 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 069K: sources/stacks-chow/tags.txt, source lines 8044–8044. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6572 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *A syntomic morphism is Gorenstein. Equivalently a flat local complete intersection morphism is Gorenstein.*

Proof. Recall that a syntomic morphism is flat and its fibres are local complete intersections over fields, see Morphisms, Lemma 01UF. Since a local complete intersection over a field is a Gorenstein scheme by Lemma 0BVA we conclude. The properties "syntomic" and "flat and local complete intersection morphism" are equivalent by More on Morphisms, Lemma 069K. □